



Fig. 3: Screenshot of the program output with graphics objects

and give correct path in MATLAB.

3. Click 'Count.' The *efy.jpg* file will be imported and counted. In our example, we have created only nine objects and therefore the program

output (refer Fig. 3) shows total objects counted as 9.0.

4. Click 'Clear' to clear the image from the screen.

You can also create a standalone

## EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on [www.efymag.com](http://www.efymag.com) website.

executable program so that you don't have to install MATLAB in order to run this program. MATLAB Compiler lets you share your MATLAB application as an executable or a shared library. Executables and libraries created with MATLAB Compiler use a runtime engine called 'MATLAB Compiler Runtime' (MCR). MCR is provided with MATLAB Compiler for distribution with your application and can be deployed royalty-free.

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## PLOTTER.M

```
function varargout = plotter(varargin)
gui_Singleton = 1;
gui_State = struct('gui_Name',       mfilename, ...
                  'gui_Singleton',   gui_Singleton, ...
                  'gui_OpeningFcn', @gui_OpeningFcn, ...
                  'gui_OutputFcn',  @plotter_OutputFcn, ...
                  'gui_LayoutFcn',  [] , ...
                  'gui_Callback',    []);
if nargin && ischar(varargin{1})
    gui_State.gui_Callback = str2func(varargin{1});
end

if nargout
    [varargout{1:nargout}] = gui_mainfcn(gui_State, varargin{:});
else
    gui_mainfcn(gui_State, varargin{:});
end
% End initialization code - DO NOT EDIT

% --- Executes just before plotter is made visible.
function plotter_OpeningFcn(hObject, eventdata, handles, varargin)
% This function has no output args, see OutputFcn.
% hObject    handle to figure
% eventdata  reserved - to be defined in a future version of MATLAB
% handles     structure with handles and user data (see GUI-DATA)
% varargin   command line arguments to plotter (see VARARGIN)

% Choose default command line output for plotter
handles.output = hObject;

% Update handles structure
guidata(hObject, handles);

% --- Outputs from this function are returned to the command
```

```
line.
function varargout = plotter_OutputFcn(hObject, eventdata, handles)
% varargout  cell array for returning output args (see VARARGOUT);
% hObject    handle to figure
% eventdata  reserved - to be defined in a future version of MATLAB
% handles     structure with handles and user data (see GUI-DATA)

% Get default command line output from handles structure
varargout{1} = handles.output;

% --- Executes on button press in pushbutton1.
function pushbutton1_Callback(hObject, eventdata, handles)
% hObject    handle to pushbutton1 (see GCBO)
% eventdata  reserved - to be defined in a future version of MATLAB
% handles     structure with handles and user data (see GUI-DATA)

a=imread('C:\efytest\efy.jpg'); %%%%%%%%%%%%%%%
%ENTER THE PICTURE LOCATION IN FILE HERE%%%%%%%%%
b=rgb2gray(a);%CONVERT RGB VALUE TO GRAY
averaging=graythresh(b);%FIND THE THRESHOLD
bw=im2bw(b,averaging);
[L,num]=bwlabel(bw,8);
detectedobjects=regionprops(L,'basic');
i=0;
for j=1:num;
if (detectedobjects(j).Area)>2500;%CHECK FOR THE COLOR WHERE IT FINDS THE INTENSITY FOR WHITE
    i=i+1;
end
end
h=ones(5,5)/25;
```